

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>

To calculate the quantity of heat required to convert a liter of water at 30°C to a liter of water at 60°C , we first note that the difference in temperature is 30°C . We multiply this with the amount of water (1 liter) and the specific heat of water , which is 1 cal/g°C . Since 1 kg = 1000 grams , we multiply this by 1000 . So the heat required is given by $1000 * 30 * 1 = 30$ Kcal .

To calculate the quantity of heat required to heat a 1 kg block of aluminum from 30°C to 60°C , we first convert this temperature difference to Celsius degrees : $60^{\circ}\text{C} - 30^{\circ}\text{C} = 30^{\circ}\text{C}$ Since the density of aluminum is not given in this question we have to assume that it 1 , so we multiply the block mass by the density * the specific heat , which are the same in this case . So the heat required is given by $1000 * 30 * .215 = 6.3$ Kcal .

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<answer> G